STEFAN FIEGUTH

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https://stefanfieguth.ca (Engineering Portfolio)

EDUCATION

UNIVERSITY OF WATERLOO Waterloo, ON
Systems Design Engineering - Bachelor's of Applied Science Sept 2023 - Apr 2028
Excellent Academic Standing
Relevant Coursework: Mechanical Design (CAD), Digital/Analog Circuits, Prototyping, Mechanics of Deformable Solids

WORK EXPERIENCE

ELECTRANS TECHNOLOGIES (Autonomous Truck Hitching Start-Up) Oakville, ON
Mechatronics Engineering Co-op Sept 2026 – Dec 2026

- Designed, implemented, and tested a new adjustable end stop for more reliable coupling cycles
- Wrote instruction manuals for field technicians to install the trailer receptacle units
- Developed/Fixed a new end-of-line tester to Q/A check all of their receptacles
- Streamlined their assembly line by designing and printing new jigs to make assembly easier
- Built new Tractor-Trailer Auto-Connecting units

WERKR TRACTORS (Electric Autonomous Tractor Start-up) Waterloo, ON
Mechanical/Electrical Engineering Co-op (3 Separate Co-op Terms) Sept 2024 – Apr 2026

- Designed, machined, assembled, and installed new mechanical parts
- Redesigned the tractor's Flame Weeding implement for early-season weeding
- Redesigned the tractor's Implement Crane to easily switch out the various implements
- Designed a new printed circuit board (PCB) to organize the main controller box, allowing for easier debugging and a more robust machine

ACCELERATED SYSTEMS INC. (High-Torque Electric Motor and Motor Controller Company) Waterloo, ON
Mechanical Engineering Co-op Jan 2024 – Apr 2024

- Helped convert gasoline and low-power electric golf carts to high-power electric golf carts
- Diagnosed and fixed broken, faulty motors; designed them to be waterproof

SILVER LAKE MENNONITE CAMP Sauble Beach, ON
Camp Counsellor, Maintenance Worker (Seasonal) 2023 – Present

- Counselled kids aged 8-16, led camp activities, cleaned cabins, refubishing/fixing buildings and toys
- Designed an automatic campfire starting system, performed chemical experiments with the kids
- Built a 3D-printed scale model of the camp

PERSONAL PROJECTS

CAMP MAP/MODEL (Gift for Silver Lake Mennonite Camp) Jan 2024 - Jun 2024

- Designed (CAD), built (woodworking/3D printing), and painted a detailed scale model of the camp
- Everything from the main buildings to small huts, bins, and signs was modelled, printed, and painted
- Included over 500 individual 3D-printed parts

PERIODIC TABLE OF ELEMENTS DISPLAY Sept 2025 - Jan 2026

- Designed (CAD), built (woodworking/3D printing), and coded (Arduino) a 4ft periodic table light-up display
- This was shown at various high schools to teach and get students excited about chemistry

WINNING BEAM COMPETITION (University of Waterloo MODs Class) Oct 2025 - Nov 2025

- Designed (CAD), built (3D printing), and tested our winning beam, having the highest strength-to-weight ratio
- Using a rigorous material selection process, finite element analysis (FEA), and our own testing equipment, we won the competition by 3D printing a beam (using Bambu PPA-CF) capable of holding 71,000 times its own weight

ADDITIONAL

Advanced Technical Skills: CAD (Onshape, SolidWorks), Machining, 3D Printing (FDM), Woodworking, CNC Milling

Developing Technical Skills: CNC Turning, Soldering, PCB Design (EDA), Marketing (Photography)

Languages: English (Fluent), German (Fluent), French (Working Proficiency)

Certifications & Training: Standard First Aid with CPR-C

Hobbies: 3D Printing, Woodworking, Soccer, Ultimate Frisbee, Pyrotechnics, Cello